

14 Rounding Applied to Set Cover

We will introduce the technique of LP-rounding by using it to design two approximation algorithms for the set cover problem, Problem 2.1. The first is a simple rounding algorithm achieving a guarantee of f , where f is the frequency of the most frequent element. The second algorithm, achieving an approximation guarantee of $O(\log n)$, illustrates the use of randomization in rounding.

Consider the polyhedron defined by feasible solutions to an LP-relaxation. For some problems, one can find special properties of extreme point solutions of this polyhedron, which can yield rounding-based algorithms. One such property is *half-integrality*, i.e., in each extreme point solution, every coordinate is 0, 1, or $1/2$. In Section 14.3 we will show that the vertex cover problem possesses this remarkable property. This directly gives a factor 2 algorithm for weighted vertex cover; namely, find an optimal extreme point solution and round all the halves to 1. A more general property, together with an enhanced rounding algorithm, called iterated rounding, is introduced in Chapter 23.

14.1 A simple rounding algorithm

A linear programming relaxation for the set cover problem is given in LP(13.2). One way of converting a solution to this linear program into an integral solution is to round up all nonzero variables to 1. It is easy to construct examples showing that this could increase the cost by a factor of $\Omega(n)$ (see Example 14.3). However, this simple algorithm does achieve the desired approximation guarantee of f (see Exercise 14.1). Let us consider a slight modification of this algorithm that is easier to prove and picks fewer sets in general:

Algorithm 14.1 (Set cover via LP-rounding)

1. Find an optimal solution to the LP-relaxation.
2. Pick all sets S for which $x_S \geq 1/f$ in this solution.

Theorem 14.2 *Algorithm 14.1 achieves an approximation factor of f for the set cover problem.*

Proof: Let $\mathcal{C}$ be the collection of picked sets. Consider an arbitrary element e . Since e is in at most f sets, one of these sets must be picked to the extent of at least $1/f$ in the fractional cover. Thus, e is covered by $\mathcal{C}$, and hence $\mathcal{C}$ is a valid set cover. The rounding process increases x_S , for each set $S \in \mathcal{C}$, by a factor of at most f . Therefore, the cost of $\mathcal{C}$ is at most f times the cost of the fractional cover, thereby proving the desired approximation guarantee. $\square$

The set cover instance arising from a vertex cover problem has $f = 2$. Therefore, Algorithm 14.1 gives a factor 2 approximation algorithm for the weighted vertex cover problem, thus matching the approximation guarantee established in Theorem 2.7.

Example 14.3 Let us give a tight example for Algorithm 14.1. For simplicity, we will view a set cover instance as a hypergraph: sets correspond to vertices and elements correspond to hyperedges (this is a generalization of the transformation that helped us view a set cover instance with each element having frequency 2 as a vertex cover instance).

Let $V_1, \dots, V_k$ be k disjoint sets of cardinality n each. The hypergraph has vertex set $V = V_1 \cup \dots \cup V_k$, and n^k hyperedges; each hyperedge picks one vertex from each V_i . In the set cover instance, elements correspond to hyperedges and sets correspond to vertices. Once again, inclusion corresponds to incidence. Each set has cost 1. Picking each set to the extent of $1/k$ gives an optimal fractional cover of cost n . Given this fractional solution, the rounding algorithm will pick all nk sets. On the other hand, picking all sets corresponding to vertices in V_1 gives a set cover of cost n . $\square$

14.2 Randomized rounding

A natural idea for rounding an optimal fractional solution is to view the fractions as probabilities, flip coins with these biases and round accordingly. Let us show how this idea leads to an $O(\log n)$ factor randomized approximation algorithm for the set cover problem.

First, we will show that each element is covered with constant probability by the sets picked by this process. Repeating this process $O(\log n)$ times, and picking a set if it is chosen in any of the iterations, we get a set cover with high probability, by a standard coupon collector argument. The expected cost of cover picked in this manner is $O(\log n) \cdot \text{OPT}_f \leq O(\log n) \cdot \text{OPT}$, where OPT_f is the cost of an optimal solution to the LP-relaxation. Applying Markov's Inequality, we convert this into a high probability statement. We provide details below.